

# NEUROSENSE: AI-DRIVEN PARKINSON'S DISEASE PREDICTION USING ONLINE HANDWRITING

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**Abstract—** *The delayed onset of symptoms and the dependence on subjective clinical evaluations, Parkinson's disease (PD) can be diagnosed only after the significant progression. In order to overcome this difficulty, we suggest an AI-driven diagnostic tool that makes use of spiral handwriting analysis -- a task that has been shown to highlight the subtle fine motor impairments linked to early-stage Parkinson's disease. In order to capture both dynamic and static handwriting characteristics, our suggested framework uses a dual-stream deep learning architecture. In order to separate online handwriting data into Beta strokes and extract sophisticated dynamic features (like pressure, velocity, and fluidity), the system first uses a Beta-elliptical model in conjunction with a fuzzy perceptual detector. To model long-term dependencies in the writing movement, a Bidirectional Long Short-Term Memory (BLSTM) network processes these temporal sequences. Convolutional Neural Networks (CNNs) are used in tandem to capture geometric irregularities that dynamic analysis alone frequently overlooks by directly extracting high-level spatial features from the spiral drawings' visual representations. To categorize subjects as either PD-positive or healthy, the outputs from the CNN and BLSTM models are combined. This technology greatly surpasses traditional techniques by successfully fusing the spatial feature extraction of a CNN with the dynamic precision of the Beta-elliptical approach. The end product is an early Parkinson's disease detection tool that is accurate, non-invasive, and computationally efficient.*

## I. INTRODUCTION

This study emphasizes spiral drawing tasks and offers a novel, non-invasive framework for online handwriting analysis- based Parkinson's disease (PD) detection. We suggest a dual- stream hybrid deep learning architecture that captures both the dynamic and static aspects of handwriting in order to overcome the shortcomings of conventional diagnostic techniques. In the first stream, the system divides online handwriting data into beta strokes using a fuzzy perceptual detector in conjunction with a beta-elliptical model. In order to model temporal motor irregularities, a Bidirectional Long Short-Term Memory (BLSTM) network processes the complex dynamic features that

are extracted by these advanced preprocessing and data augmentation steps.

Key Features:

- **Online Handwriting-Based Detection** Detects Parkinson's Disease using real-time digital handwriting input captured via a graphics tablet.
- **Advanced Feature Extraction** Utilizes the Beta-Elliptical approach to capture both kinematic (velocity) and geometric (stroke shape) details.
- **Fuzzy Perceptual Detector** is integrated to extract uncertainty-tolerant visual perception features.
- **Smart Stroke Segmentation** - Divides handwriting into Beta strokes based on the change in the person's velocity, thus enabling a detailed

analysis of the motor skills.

- **Data Augmentation & Preprocessing** - Geometric transformations are used to increase the variety of the dataset.

## II. PURPOSE OF THE PROJECT

It is still a big problem to figure out the early-stage Parkinson's Disease. This is because subtle motor impairments are easy to miss and people often wrongly attribute them to other causes. The traditional ways of diagnosis take a lot of time, need special clinical settings, and may not be available to everyone, particularly in low-resource areas. As a result, treatment is delayed, and the condition of patients as well as the healthcare systems gets worsened. Most of the existing diagnostic methods have not yet fully realized the potential of handwriting as an easily accessible, real-time, and non-invasive biomarker. The patients and doctors cannot rely on the subjective evaluation that data-driven methods are lacking; thus, they are at variance with one another. This research is aimed at removing these obstacles by creating an AI-powered Parkinson's detection system that incorporates online handwriting analysis, machine learning, and deep temporal modelling.

The innovative system will be using the Beta-Elliptical model and Fuzzy Perceptual Detectors to identify and extract fine-grained neuromotor features and SPD networks to classify them, in a highly intelligent manner, from the raw handwritten input. This is a diagnosed support accessible, language-aware, and user-friendly tool, which is clinically and naturally usable. It is the essence of early detection, personalized monitoring, and timely intervention changing radically the concept of the digital era-based neurological diseases like Parkinson's. A detection system that employs deep temporal modeling, machine learning, and online handwriting analysis.

The proposed system will use the Beta-Elliptical model and fuzzy perceptual detectors to smartly extract fine-grained neuromotor features that will be classified through bidirectional LSTM networks. It provides a simple, linguistically informed, and clinically as well as practically-designed diagnostic support tool. The goal is to change the comprehension and therapy of neurological disorders such as Parkinson's in the Digital Era by facilitating early detection, personalized monitoring, and timely intervention.

## III. SCOPE OF THE PROJECT

The goal of this project is to develop a sophisticated, artificial intelligence (AI)-driven

system that can identify early indicators of Parkinson's disease by analyzing online handwriting. By utilizing sophisticated methods like the Bidirectional LSTM for classification and the BetaElliptical approach for feature extraction, the system is able to precisely detect neuromotor abnormalities associated with Parkinson's disease. The procedure is non-invasive, economical, and real-time

Here's what the platform will cover:

- **Real-Time Detection:** Instantly recognizes early Parkinson's disease symptoms from online handwriting input.
- **AI-Based Analysis:** This method looks for neuromotor irregularities in handwriting using trained machine learning models.
- **Fuzzy Perceptual Detection:** The process involves detecting uncertain or very subtle characteristics of the handwriting that are hidden using feature extraction based on fuzzy logic.
- **Stroke Segmentation:** Breaks handwriting into Beta strokes for precise motion tracking and evaluation.
- **Non-Invasive & Fast:** The method of diagnosing Parkinson's disease early is simple, without any direct contact, and quick.
- **Personalized Assessment:** Adjusts to different writing styles to guarantee precise, user-specific outcomes.

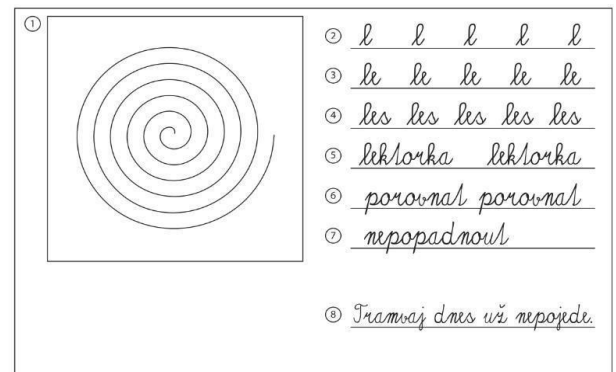


Fig. 1. HANDWRITING TECHNIQUES AND SPIRAL TRACING FOR MOTOR IMPAIRMENT

## IV. PROBLEM STATEMENT

Parkinson's Disease (PD) is a progressive neurodegenerative condition that impairs the brain's ability to regulate motor functions. The major symptoms include tremors, bradykinesia, muscle stiffness, balance problems, and micrographia. As the disease progresses, it becomes increasingly difficult for the affected person to carry out daily

routes.

Almost indiscernible early signs are one of the biggest hurdles associated with Parkinson's disease. At first, the symptoms might be so faint that they could be mistaken for the effects of aging or some other health condition. Subsequently, due to this factor, diagnosis is often made at an advanced stage when the disease has caused considerable damage.

About 80% of Parkinson's cases are confirmed through the evaluation of symptoms in a physical examination by the doctor. Unfortunately, this method lacks precision since it is totally dependent on the expertise of the physician and can differ from one doctor to another. Besides, such investigations require some time and might not be accessible in every hospital or clinic, particularly in far-off or less-privileged localities.

## V. OBJECTIVES

- Detect early signs of Parkinson's Disease using handwriting analysis.
- Track disease progression and provide actionable insights through an AI-powered chatbot.
- Transform Parkinson's care by integrating advanced technology into diagnosis and support.
- Help users seek early consultation and maintain regular checkups with engaging chatbot reminders.

## VI. METHODOLOGY

- *Data Collection and Augmentation* To train and

rotation were used to replicate the variability of natural handwriting in order to maximize the Convolutional Neural Network (CNN).

- This increases the variety of training, guaranteeing that the model effectively generalizes across subjects and captures important spatial features.

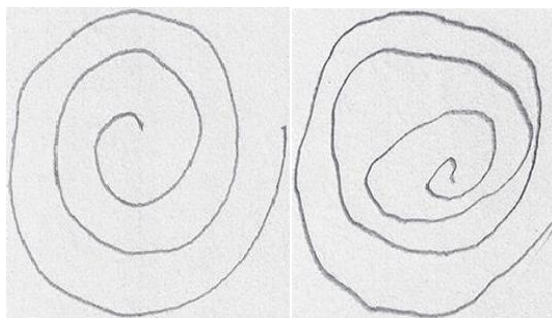


Fig. 2. Healthy and Parkinson's Spiral Patterns in

## Handwriting Assessment

### A. Preprocessing and Noise Filtering

- To guarantee consistency and quality, collected handwriting data is first put through preprocessing procedures.
- Outliers and noise are eliminated by filtering the handwriting signals, which are recorded as a series of pen positions over time. The data is then standardized by applying normalization, which aligns variances caused by writing speed, size, and device settings.
- In order to guarantee that the feature extraction and classification procedures that follow operate on clear, comparable data, these preprocessing steps are essential.

### A. Beta Stroke Segmentation

- The system uses a stroke segmentation method called beta stroke segmentation, which is based on variations in pen velocity.
- A more detailed examination of motor control is made possible by breaking down each handwriting sample into smaller, meaningful motion components (also known as "beta strokes").
- This is especially helpful for capturing the slowness, hesitation, and tremors that are characteristic of Parkinsonian handwriting. Segmentation enhances the quality of features.

### A. Feature Extraction via Beta-Elliptical and Fuzzy Perceptual Models

- Two sophisticated feature extraction techniques are used to analyze handwriting segments: Each stroke's kinematic (velocity, acceleration) and geometric (shape, curvature) characteristics are extracted by the beta-elliptical model.
- This offers comprehensive information about the motor function that underlies handwriting. Fuzzy Perceptual Detector: Models how humans see handwriting using fuzzy logic.
- It records subtle characteristics and irregularities that may be difficult to measure with conventional models, such as variations in stroke smoothness or hesitation.

### A. Classification with Bidirectional LSTM

- A Bidirectional Long Short-Term Memory (BLSTM) neural network receives the extracted features as input for classification. Because of its ability to accurately model temporal dependencies and context from both historical and upcoming input sequences, BLSTM was selected.
- This makes it particularly useful for identifying patterns and changes in neuromotor impairment over time in handwriting strokes.
- A prediction about whether the handwriting sample

exhibits symptoms of Parkinson's disease is produced by the model.

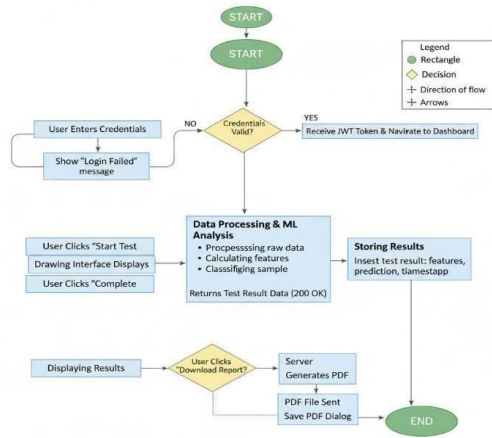


Fig. 3. Flow Diagram

1. **Integrated Pipeline:** A complete pipeline is implemented, covering handwriting input, authentication, feature extraction, ML inference, and report generation.

2. **Secure Processing:** All requests use JWT-based authentication to ensure secure data transmission and access.

3. **Real-time Analysis:** The extracted handwriting features are classified and interpreted by the machine learning model in real time.

4. **Test results:** Stored in the database for later longitudinal analysis.

5. **Instant Reporting:** A structured PDF report is generated automatically and made available for immediate download.

## VII. RESULTS AND DISCUSSION

### A. Performance of Feature-Engineered Classification Models

The eight stylus-based tasks (one spiral and seven alphabets) of 38 healthy and 37 PD subjects were utilized for a dataset. The five attributes—x, y, pressure, button state, and timestamp—constituted the raw input. Six features—velocity, acceleration, total time, pressure, button-state

transitions, and task type—were extracted and used to train five classifiers: logistic regression, SVM, random forest, gradient boosting, and k-NN. Out of these, the Random Forest model was able to identify the correct class with the highest accuracy and the best overall classification metrics. To determine the condition of the user, their sketch is therefore transformed into the same six features, and the Random Forest model is then given these features to return the prediction of Healthy or PD.

### B. Performance of CNN Model Using Spiral-Drawing Images

A CNN-based classifier was developed using a spiral-drawing image dataset with 126 healthy and 126 PD samples for training and 36 healthy and 36 PD samples for testing. Five backbone architectures, namely MobileNetV2, ResNet, EfficientNet-B0, DenseNet, and Xception, were compared.

The EfficientNet-B0 model with the highest performance and the best overall classification metrics across all architectures was able to make a clear distinction between spirals of healthy and PD subjects. A spiral image is uploaded by the users, and the CNN, along with Grad-CAM, which is used to visualize the areas that are most influential to the prediction, classifies it as healthy or PD.

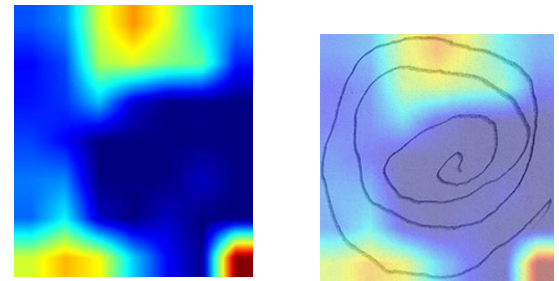


Fig. 4. Heatmap and Grad-CAM Visualization for PD Prediction.

## CONCLUSION

AI-driven handwriting analysis provides a dependable, non-invasive, and economical approach for the early detection of Parkinson's disease (PD), as the NeuroSense project shows. The system successfully separates handwriting irregularities associated with Parkinson's disease (PD) symptoms like tremors, bradykinesia, and micrographia by incorporating machine learning techniques like the Beta-Elliptical model. It is clear from the cited case study that handwriting parameters like pressure, speed, in-air movement, and entropy-based features have significant diagnostic value and can classify PD with almost greater accuracy. By combining these discoveries with NeuroSense's neural network methodology, diagnostic accuracy is further improved, and real-time handwriting

assessment via digital tablets is made possible. Overall, this project demonstrates that handwriting can serve as a biomarker for Parkinson's disease detection, providing medical professionals with an effective and scalable automated decision-support tool. It provides quicker, data-driven, and easily accessible Parkinson's screening by bridging the gap between medical science and artificial intelligence.

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